

XRPL Validator — Daily Engineering Report • 2026-08-07

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **8 commits** (xrpl-validator 8)
- 1 test / vector commit

Engineering changes

- **feat(xrpl-ledger): DX_ULP — the 1-ulp upper bound is unobservable, and the note was backwards** (f2d9d9a , xrpl-validator) Hunted for a ledger where the composed upper bound's last digit decides an admission, per the open item in 005ad9a. There isn't one.
- **fix(xrpl-node): the FFI oracle replayed every old fixture under TODAY'S rules** (a191b79 , xrpl-validator) Re-minting the stale freshfinds/105663160 fixture (it lacked tx_json and had been skipped by every gate since 24 July) surfaced this.
- **fix(xrpl-node): prefetch the NFT offer's OWNER — the accept names it by hash** (60fb653 , xrpl-validator) Root cause of #105663160 5CCAAE138F560F07 , and it is OURS, not rippled's.
- **refactor(xrpl-ledger): remove DX_SEL — it now compares an abandoned model** (17a99ef , xrpl-validator) DX_SEL was added in dbb9110 to find a ledger where our estimate-based strand selection differed from rippled's upper-bound one. It found the answer, the answer was acted on, and the detector outlived its premise: since 03c2cb9 the engine SELECTS by that upper-bound model (order), so DX_SEL was comparing an abandoned model against the live one and reporting "mismatches" that described nothing real.
- **fix(xrpl-ledger): destination checks come before the funding check** (5626357 , xrpl-validator) #106136429 6FF692D40C4F , from the 2026-08-07 fresh batch. The FFI leg was clean, so the pre-state was sound and the engine was wrong.
- **fix(xrpl-ledger): a payment crosses the payer's own offer** (8d8ca2d , xrpl-validator) #106137720 F2C989D4843A , from the 2026-08-07 fresh batch. FFI leg clean, so the pre-state was sound and the engine was wrong.
- **feat(xrpl-ledger): max_offer — rippled's single-path AMM slice, trace-exact** (a171405 , xrpl-validator) Groundwork for the AMM bridge slice-sizing family, with the two specimens from the 2026-08-07 batch. NOT wired in; see below.

Tests and campaigns

- **test(xrpl-ledger): the 3 red regression tests encoded a 10 XRP reserve** (751d5d4 , xrpl-validator)

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Validator Bond, Slashing, and Reputation Score** (consensus) Add a new owner-reserved ledger object `ltVALIDATOR_BOND` (`keylet::validatorBond(masterPubKey)`) holding `sfBondAmount`, `sfValidatorPublicKey`, `sfReputation`
- **Reputation-Discounted Collateral for On-Ledger Loans** (lending) Add a protocol-maintained ledger object `ltLOAN_CREDIT_SCORE` (`keylet::loanCreditScore(borrower)`, no owner reserve) tracking `sfOnTimeRepayments`, `sfDefau`
- **Delegated Compliance Approval for Restricted Transfers** (rwa) Extend XLS-75 Permission Delegation's `DelegateSet` transaction with a new granular permission code `PermissionTransferApproval`, scoped by the issuer to

Appendix — commits

Time	Repo	Commit	Subject
01:10	xrpl-validator	f2d9d9a	feat(xrpl-ledger): DX_ULP — the 1-ulp upper bound is unobservable, and the note was backwards
09:46	xrpl-validator	751d5d4	test(xrpl-ledger): the 3 red regression tests encoded a 10 XRP reserve
10:53	xrpl-validator	a191b79	fix(xrpl-node): the FFI oracle replayed every old fixture under TODAY'S rules
12:13	xrpl-validator	60fb653	fix(xrpl-node): prefetch the NFT offer's OWNER — the accept names it by hash
14:36	xrpl-validator	17a99ef	refactor(xrpl-ledger): remove DX_SEL — it now compares an abandoned model
17:17	xrpl-validator	5626357	fix(xrpl-ledger): destination checks come before the funding check
18:00	xrpl-validator	8d8ca2d	fix(xrpl-ledger): a payment crosses the payer's own offer
23:08	xrpl-validator	a171405	feat(xrpl-ledger): max_offer — rippled's single-path AMM slice, trace-exact